INCH-POUND
MIL-DTL-24640/16E
8 November 2011
SUPERSEDING
MIL-DTL-24640/16D
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, -20 TO +105 °C, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, 18 AWG, TYPE 2XSW (INCLUDING VARIATION TYPE 2XSOW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction (Watertight)

First	-	Tin-coated copper conductor in accordance with table I.
Second	-	Insulation material and basic construction requirements shall be in accordance with SAE-AS81044/12 and tested in accordance with the component wire ratings and requirements listed on this specification sheet. Conformance testing shall be in accordance with SAE-AS81044 prior to cabling.
Third	-	Two conductors, one black and one white, twisted together. The pitch shall be 1±¼ inch.
Fourth	-	Optional binder tape.
Fifth	-	Braided shield of 38 AWG tin-coated copper strands. Braid coverage shall be no less than 90 percent determined in accordance with WC-27500.
Sixth	-	Shield insulation of two polyester tapes or polyester tape and extruded jacket. Identification codes applied using Method 2. (NOTE: This step is not required for Type 2XSW1.)
Seventh	-	The required number of shielded, twisted pairs shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used, as necessary, to form a round cable.
Eighth	-	Optional binder tape.

Construction Variant for Type 2XSW

Ninth	-	Cross-linked polyolefin jacket (see table II for thickness). NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

Construction Variant for Type 2XSOW

Ninth	-	Braided shield, tin-coated copper strands. Braid must be constructed to meet surface transfer impedance requirements as listed herein.
Tenth	-	Optional binder tape.

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Eleventh - Cross-linked polyolefin jacket (see table III for thickness).
NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

TABLE I. Component wire details.

Watertight conductor		Finished wire		
Wire size (AWG)	Stranding (no. x AWG)	Test load life cycle accelerated aging tests (lbs ±3%)	Low temp. at -40 °C	
			Mandrel diameter (inches)	Test load (lbs)
18	7 x 26	0.250	1.0	1.0

TABLE II. Construction details.

PIN M24640/16	Type number	Number of pairs	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01UN	2XSW-1	1	0.034	0.240	0.258	52.9
-02UN	2XSW-3	3	0.041	0.436	0.470	175.0
-03UN	2XSW-7	7	0.044	0.573	0.617	330.0

TABLE III. Construction details - overall shield.

PIN M24640/16	Type number	Number of pairs	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01UO	2XSOW-3	3	0.041	0.487	0.525	227.0
-02UO	2XSOW-7	7	0.044	0.608	0.656	401.0
-03UO	2XSOW-12	12	0.047	0.802	0.864	665.0
-04UO	2XSOW-19	19	0.050	0.938	1.010	950.0
-05UO	2XSOW-30	30	0.057	1.180	1.270	1442.0

REQUIREMENTS:

Qualification required.

Construction details: See tables I, II, and III.

Component wire: The component wire shall be in accordance with SAE-AS81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating	135 °C (275 °F) (max.) conductor temperature
Voltage rating	600 volts (root mean square) at sea level
Accelerated aging	Oven temperature, 250±2 °C (482±3.6 °F) for 6 hours
Low temp. cold bend	-40 °C (-40 °F) for 4 hours

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Blocking	150±2 °C (302±3.6 °F)
Color	In accordance with MIL-STD-104, Class 1; white preferred
Flammability	30 seconds (max.)
	3.0 inches (max.)
	No flaming of tissue paper
Humidity resistance	5000 megohms for 1000 feet (min.) insulation resistance after humidity exposure
Dielectric test	
Primary insulation (when tested in lieu of spark test)	6.0 kilovolts (peak), 100 percent impulse test
Finished wire	8.0 kilovolts (peak), or 3 kHz spark test 5.5 kV, 100 percent test
Insulation resistance	5000 megohms for 1000 feet (min.)
Life cycle	Oven temperature, 200±2 °C (392±3.6 °F) for 120 hours
Physical properties of primary insulation	
Tensile strength	2500 lb/in^2 (min.)
Elongation	150 percent (min.)
Shrinkage	0.125 inch (max.) at 250±2 °C (482±2 °F)
Spark test primary insulation	1500 volts (root mean square), 3 kHz, 100 percent test
Surface resistance	500 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F)
Maximum change in measurement	0.060 inch
Wet dielectric test	2500 volts (root mean square)
Wrap test	Mandrel wrap test required

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	7.11
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor and shield	1000
Component shield to component shield	200
Component shield to overall shield (Type 2XSOW only)	500
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor and shield	2500
Shield to shield (component shield only)	100

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Shield to shield (component shield to overall shield, Type 2XSOW only)	100
Conductor and shield continuity	No failure
Jacket flaws (Type 2XSOW only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance	
Mutual capacitance (at 1 MHz, picofarads/foot (pF/ft), max.)	60
Capacitance unbalance (percent, max.)	15
Characteristic impedance (at 1 MHz, ohms)	30 (min.), 53 (max.)
Watertightness (see table IX of MIL-DTL-24640 for limits of water leakage)	No failure
Weight	No failure
Group B:	
Cold bending	No failure
Drip	No failure
Cross-link proof test (percent, max.)	
Jacket	50
Tear (lb/in thickness, min.)	35
Physicals (unaged)	
Jacket	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Attenuation (at 3 MHz, dB/100 feet, max.)	Not applicable
Gas flame (1 hour)	Not applicable
Group C:	
Physicals (aged), air oven	
Jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (conductor insulation, cycles, min.)	Not applicable
Permanence of printing (jacket, cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable sealant removability	No failure

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Shrinkage	No failure
Surface transfer impedance (Type 2XSOW only)	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60

Group D:

Flame propagation	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	10
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Component wire	No failure

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UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)	
2XSW-1, 2XSW-3, 2XSW-7, 2XSOW-3, 2XSOW-7, 2XSOW-12, 2XSOW-19, 2XSOW-30	3 - 12	500 feet nominal
	19 & 30	250 feet nominal

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-084)

Review Activities:
Army – AR, AV, CR, CR4
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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